Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering

Course Code & Title: CB3491 Cryptography and Cyber Security

Name of the Faculty member: Mrs.P.Sabeena Burvin, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit I /** OSI Security Architecture
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** Think-Pair-Share
- **Date of Implementation:** 02.08.2025
- **Justification:**

  The Think–Pair–Share (TPS) strategy encourages students to engage deeply with a concept through individual reflection, peer discussion, and collective sharing. In this session, students explored the OSI Security Architecture and its security services, which are essential for protecting data during transmission across network layers.

  The objective was to understand how these services work together to ensure confidentiality, integrity, authentication, and non-repudiation in communication systems.

Time Allotted for the Activity: 30 minutes

Details of the Implementation:

Objectives:

- To understand the purpose and structure of the OSI Security Architecture.
- To identify and describe various **OSI security services**.
- To analyze real-world applications of these services in securing networks.
- To promote collaborative learning through discussion and sharing of perspectives.

Think Phase (Individual Work)

Students individually reviewed materials and reflected on the following questions:

1. What is the purpose of the **OSI Security Architecture**?
2. What are the different **security services** defined in it?
3. How do these services ensure secure data communication?

Individual Notes:

- The **OSI Security Architecture**, developed by ISO, provides a systematic framework for implementing and managing network security.
- It defines **security attacks**, **security services**, and **security mechanisms**.
- **Main Security Services:**
    1. **Authentication** – verifies the identity of communicating entities.
    2. **Access Control** – restricts unauthorized access to resources.
    3. **Data Confidentiality** – ensures that information is not disclosed to unauthorized users.
    4. **Data Integrity** – protects data from unauthorized modification.
    5. **Non-Repudiation** – ensures that a sender or receiver cannot deny having sent or received data.

Pair Phase (Collaborative Discussion)

Students paired up to compare their understandings and discuss practical examples of each service.

Key Discussion Points:

- **Authentication:** Passwords, digital certificates, and biometrics in login systems.
- **Access Control:** Role-based access in operating systems and databases.
- **Data Confidentiality:** Use of encryption (DES, AES, SSL/TLS).
- **Data Integrity:** Checksums, hash functions (SHA, MD5).
- **Non-Repudiation:** Digital signatures and block chain transaction logs.

Insights from Pair Discussions:

- All services are interrelated; for example, encryption contributes to both confidentiality and integrity.
- Implementation depends on the application layer and user requirements.
- Modern networks combine these services using **cryptographic protocols** like HTTPS and IPSec.

Share Phase (Class Discussion)

Each pair presented their key findings to the class. The instructor summarized and connected student insights to the OSI Security Architecture model.

Highlights from the Class Discussion:

- The OSI model provides a **conceptual framework**; it does not prescribe specific technologies.
- Security services operate across multiple layers—for example, **authentication** at the session layer and **confidentiality** at the presentation layer.
- **Real-world applications:** VPNs, secure email (PGP), and online banking use these services simultaneously.
- Emphasis on the balance between **security and performance** in system design.

Learning Outcomes

By the end of the session, students were able to:

- Explain the components of the OSI Security Architecture.
- Identify and describe the five primary OSI security services.
- Apply theoretical understanding to practical network security scenarios.
- Collaborate effectively and articulate security concepts clearly

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO12
CO1	3	2	2	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO Mappe

Innovative Practice	PO1	PO2	PO3	PO4	PO8	PO9	PO12
	3	2	2	2	1	1	1
	Basic Computer Science Engineering and mathematics Knowledge for identifying various attacks.	Analyzing various Security mechanisms are highly correlated.	Investigate various aspects of Services provided.	Use of virtual environment for identifying various attacks.	Students will complete the assignments / tasks in an ethical way.	The students will work effectively as an individual and team for solving the identification process.	Updating the concepts to the current trends are moderately correlated

- **Images / Screenshot of the practice:**

Students Think and Share their Thoughts and Topic Presentation

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Working as a pair, they can share their knowledge with them and come up

with better understanding about the topic.

- o Different types of reasoning/justification based questions are discussed in the class room, so students come to know how to answer/draw diagrams whenever new real time system is asked in the question

❖ ***Benefit of the practice:***

- o Students are actively engaged.
- o Students learn from each other.
- o Builds a friendly, yet academic atmosphere.
- o Includes all the students in the teaching-learning process.

❖ ***Challenges faced in implementation:***

- Students felt very difficult in this activity and could not complete in time whenever new real time system asked in the question.
- Students are hesitating to speak in front of all.
- It took a longer time than other strategies.

References:

❖ https://www.ritrjpm.ac.in/images/computer-science/2023-2024/IP/TPS_CCS_KJK_23-24.pdf

❖ https://www.readingrockets.org/strategies/think-pair-share

❖ https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improveyour-classroom-discussions1704.